Notes on: Advances in Dynamical Low-Rank Approximations for random PDEs and Data Assimilation

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June 23, 2026

Problem

1. (a) Monte Carlo is non-intrusive and dimension-independent, but the error decays only as $\mathscr{O}(N_s^{-1/2})$.
   (b) Stochastic collocation and stochastic Galerkin methods both suffer from the curse of dimensionality.
2. Reduced Basis methods has to choose between the low-dimensionality and expressivity of chosen type of basis.

Key idea

1. Reduce the dimensionality of the problem by projecting the problem to a much smaller linear subspace.
2. Opt for separate space and stochastic bassis of which both depend on time.

What I'd prove/extend